

GATE 2027

Computer Science (CS) + Data Science & AI (DA)

Complete guide: dates, pattern, syllabus, overlap, a 19-week plan for a working engineer, and exam strategy.

At a glance	
Conducting body	IIT Madras (GATE 2027)
Exam dates	6, 7, 13, 14, 20, 21 February 2027 (forenoon and afternoon sessions)
Format	Online computer-based test, 3 hours, 65 questions, 100 marks
Results (expected)	About 19 March 2027
Score validity	3 years from result date
Your papers	CS and DA (a permitted two-paper combination)
Days left (from 28 Sep 2026)	About 131 days to 6 Feb 2027 - roughly 19 weeks

Verify on the official portal. Dates and pattern here are drawn from IIT Madras-based coverage on third-party sites. Your exact exam day, session and city come only with the admit card. Syllabus PDFs: gate2027.iitm.ac.in (CS and DA syllabus documents).

1. Key dates and process

Item	Detail
Registration (GOAPS)	Opened 2 Sep 2026; regular deadline 27 Sep 2026; extended with late fee to 5 Oct 2026
Exam days	6-7, 13-14, 20-21 Feb 2027. CS and DA fall on separate slots when you take both
Result	Expected about 19 Mar 2027
Scorecard	Free download for a limited period; later access available for a fee (about Rs. 500 per paper, until 31 Dec 2027, per coverage of the brochure)
Score validity	3 years from result announcement

What normally happens between now and the exam

- **Application data correction window** (usually October): re-check name spelling, photo, signature, category, paper codes, and exam-city choices.
- **Admit card** (usually early January): confirms date, session, city and centre for each paper. Carry the original photo ID exactly as named in the brochure.
- **Mock/virtual calculator practice**: the exam provides an on-screen calculator; personal calculators are not allowed.
- **Response sheet and answer key** follow the exam; a challenge window comes before the final result.

The bullet list above reflects the usual GATE cycle. Exact dates for correction, admit card and answer-key windows are announced on the official portal - track them there.

2. Eligibility

- Third year or higher of a government-approved undergraduate program (Engineering, Technology, Architecture, Science, Commerce, Arts, Humanities), or already completed one.
- **No age limit** and **no limit on attempts**.
- Your B.E. in Computer Engineering qualifies you for both CS and DA.
- Keep your category certificate and other documents ready in case the organizing institute asks for verification; follow the brochure's rules on certificate validity.

3. Exam pattern and marking

Feature	GATE CS	GATE DA
Duration / mode	3 hours, online CBT	3 hours, online CBT
Questions / marks	65 questions, 100 marks	65 questions, 100 marks
General Aptitude (GA)	10 questions, 15 marks	10 questions, 15 marks
Core subject	55 questions, 85 marks	55 questions, 85 marks
Question types	MCQ, MSQ, NAT	MCQ, MSQ, NAT

Section	1-mark questions	2-mark questions	Total
General Aptitude	5	5	15 marks
Core subject	25	30	85 marks

Negative marking: only for wrong MCQs - minus 1/3 on a 1-mark MCQ, minus 2/3 on a 2-mark MCQ. **MSQ** and **NAT** carry no negative marking, and MSQ needs all correct options selected with no wrong option.

Guessing maths

A blind guess on a 4-option MCQ has expected value zero (1/4 chance of +1, 3/4 chance of minus 1/3). Once you can eliminate even one option, guessing has positive expected value. NAT and MSQ are the safest place to attempt everything, but a reasoned answer beats a blank.

4. GATE CS syllabus (85 marks core + 15 GA)

Approximate weightage below comes from general past-paper trends, not from IIT Madras; treat it as a planning guide only. Coverage notes reflect a working C++/Linux background.

Subject	Topics (per official syllabus)	Approx. marks	Your edge
Engineering Mathematics	Discrete Math (logic, sets, relations, functions, partial orders and lattices, groups, graph connectivity/matching/coloring, counting, recurrences, generating functions); Linear Algebra (matrices, determinants, linear systems, eigenvalues, LU); Calculus (limits, continuity, maxima/minima, MVT, integration); Probability and Statistics (random variables, uniform/normal/exponential/Poisson/binomial, Bayes)	10-15	Shared with DA - study once
Digital Logic	Boolean algebra and minimization (K-map, tabular), combinational and sequential circuits, number representation, fixed/floating-point arithmetic	5-7	Bit-level familiarity helps
Computer Organization and Architecture	Instruction set and addressing modes, ALU design, hardwired and microprogrammed control, I/O (interrupt and DMA), instruction pipelining and hazards; memory hierarchy and cache are commonly asked	6-9	Cache/memory intuition from systems work
Programming and Data Structures	C programming, recursion, arrays, stacks, queues, linked lists, trees, BSTs, binary heaps, graphs	6-9	Strong - but trace C output by hand (pointers, undefined behaviour, precedence)
Algorithms	Searching, sorting, hashing, asymptotic analysis, greedy, dynamic programming, divide-and-conquer, graph traversals, MST, shortest paths	8-11	Reinforces your DSA practice
Theory of Computation	Regular expressions and finite automata, CFG and PDA, regular/CFL properties, pumping lemma, Turing machines, undecidability	6-8	Needs fresh study for most engineers
Compiler Design	Lexical analysis, parsing, syntax-directed translation, runtime environments, intermediate code, local optimization, data flow analysis (constant propagation, liveness, common subexpression elimination)	4-6	Build/toolchain awareness helps
Operating Systems	System calls, processes, threads, IPC, concurrency and synchronization, deadlock, CPU and I/O scheduling, memory management, virtual memory, file systems	7-10	Very strong - daily Linux work
Databases	ER model, relational algebra, tuple calculus, SQL, integrity constraints, normal forms, file organization, B and B+ trees, transactions and concurrency control	6-9	Shared with DA (partly)

Subject	Topics (per official syllabus)	Approx. marks	Your edge
Computer Networks	OSI and TCP/IP layering, switching, data link (error detection, MAC, Ethernet), distance-vector and link-state routing, IPv4, fragmentation, CIDR, NAT, flow and congestion control, TCP, socket API, DNS, HTTP	6-9	Sockets/TCP practical exposure

General Aptitude (15 marks)

- **Verbal:** grammar, vocabulary, reading comprehension, sentence ordering.
- **Quantitative:** data interpretation, ratio, percentages, powers, logarithms, permutations and combinations, series, mensuration, geometry, elementary statistics and probability.
- **Analytical:** deduction and induction, analogy, numerical relations.
- **Spatial:** translation, rotation, scaling, mirroring, assembling, paper folding and cutting, 2D/3D patterns.
- Target 13 to 15 out of 15 - this is the most reliable score in the paper. Practice about 30 minutes a day.

5. GATE DA syllabus (85 marks core + 15 GA)

DA has seven core sections. It trades the breadth of CS for depth in mathematics and machine learning - probability, linear algebra and ML tend to decide ranks.

Section	Topics (per official syllabus)	Priority
1. Probability and Statistics	Counting; probability axioms, sample space, events, independence, mutually exclusive events; marginal, conditional and joint probability; Bayes theorem; conditional expectation and variance; mean, median, mode, standard deviation; correlation and covariance; discrete random variables and PMF; uniform, Bernoulli, binomial; continuous random variables and PDF/CDF; uniform, exponential, Poisson, normal, standard normal, t and chi-squared distributions; conditional PDF; central limit theorem; confidence interval; z-test, t-test, chi-squared test	Very high
2. Linear Algebra	Vector space, subspaces, linear dependence and independence, matrices, projection, orthogonal, idempotent and partitioned matrices, quadratic forms, systems of linear equations, Gaussian elimination, eigenvalues and eigenvectors, determinant, rank, nullity, projections, LU decomposition, singular value decomposition	Very high
3. Calculus and Optimization	Functions of a single variable, limits, continuity, differentiability, Taylor series, maxima and minima, optimization involving a single variable	Medium
4. Programming, DSA	Python programming; stacks, queues, linked lists, trees, hash tables; linear and binary search; selection, bubble and insertion sort; merge sort and quicksort; intro to graph theory, traversals and shortest path	High (easy marks if DSA is strong)
5. Database Management and Warehousing	ER model, relational model, relational algebra, tuple calculus, SQL, integrity constraints, normal forms, file organization, indexing, data types, normalization, discretization, sampling, compression, data warehouse modeling	Medium
6. Machine Learning	Supervised learning: simple and multiple linear regression, ridge regression, logistic regression, k-NN, Naive Bayes, LDA, SVM, decision trees, bias-variance trade-off, cross-validation (LOO, k-fold), multi-layer perceptron and feed-forward neural network. Unsupervised: k-means and k-medoid, hierarchical clustering (single and multiple linkage), dimensionality reduction, PCA	Very high

Section	Topics (per official syllabus)	Priority
7. Artificial Intelligence	Search (informed, uninformed, adversarial); logic (propositional, predicate); reasoning under uncertainty: conditional independence representation, exact inference by variable elimination, approximate inference by sampling	High

6. CS and DA overlap - where to save effort

Area	Shared?	How to handle
General Aptitude	Fully shared	One daily habit covers both papers
Linear Algebra	Shared, DA goes deeper	Study to DA depth (SVD, projections, quadratic forms) once
Probability and Statistics	Shared, DA goes much deeper	Study to DA depth (distributions, tests, CLT) once
Calculus	Mostly shared	DA adds Taylor series and single-variable optimization
Programming, DS and Algorithms	Overlapping	CS uses C, DA uses Python; searching, sorting, trees, graph basics are common
Databases	Partly	ER, relational algebra, SQL, normal forms, indexing common; DA adds warehousing
OS, Networks, COA, Digital, Compiler, TOC	CS only	Pure CS-paper time
Machine Learning, AI	DA only	Pure DA-paper time; builds on the shared maths

Practical rule: build one shared maths and aptitude track first, then run two separate PYQ and mock tracks for the papers' different cores, with separate weak-topic lists.

7. A 19-week plan for a full-time engineer (28 Sep 2026 to 6 Feb 2027)

Assumes about 20 to 25 hours a week (roughly 2 hours on weekdays, 5 to 7 hours per weekend day). Scale the plan to your real schedule, but keep the order.

Phase	Dates	Focus	Output
0. Setup	28 Sep - 11 Oct (2 wks)	Diagnostic set from both syllabi; download official syllabus PDFs and PYQs; pick one book per subject; start daily GA (30 min); put C++ roadmap on maintenance	Baseline scores; ranked list of weak subjects; fixed weekly timetable
1. Core build	12 Oct - 22 Nov (6 wks)	Shared maths (Discrete, Linear Algebra, Probability, Calculus) + high-yield CS: Algorithms, DS/PDS, OS, Databases, Networks. Topic-wise PYQs after each topic	Notes and formula sheet per subject; PYQ accuracy tracked by topic
2. Complete coverage	23 Nov - 27 Dec (5 wks)	CS: TOC, COA, Digital Logic, Compiler. DA: Machine Learning, AI, Statistics tests, warehousing, Python refresher if needed	Every syllabus topic touched at least once; error log started
3. Mocks and revision	28 Dec - 24 Jan (4 wks)	Full-length mocks alternating CS and DA (about 2 per week), deep review of every mock, revision of error log and formula sheets, PYQ full papers	Timed strategy fixed; weak topics patched

Phase	Dates	Focus	Output
4. Taper	25 Jan - exam (about 2 wks)	Light revision, formula sheets, one mock per paper under exam timing, sleep and routine; confirm admit card, route and ID	Calm, rested exam day

Weekly rhythm

- **Weekdays:** one focused block on a subject (concept plus 15 to 25 PYQs) and a short daily GA block.
- **Saturday:** longer topic study plus a sectional test.
- **Sunday:** weekly review - redo wrong questions, update the error log, plan next week. From Phase 3 use Sunday for a full mock.
- **Your C++ roadmap:** it runs to about April 2027, overlapping GATE. Reduce it to a light maintenance slot (for example 30 to 45 minutes a few days a week, using DSA problems that also serve the Algorithms section) and resume fully after the last paper.

Other exams matter too: if you have other exam dates in the same window, map them on the same calendar now and protect the mock-test weeks in December and January.

8. Resources

Commonly used reference books - pick one per subject rather than collecting all of them.

Subject	Common references
Algorithms and DS	Cormen et al. (CLRS); Sahni-Horowitz; Weiss or Tanenbaum for DS
Operating Systems	Silberschatz-Galvin-Gagne; Tanenbaum
Computer Networks	Kurose-Ross; Forouzan; Tanenbaum
Databases	Silberschatz-Korth-Sudarshan; Ramakrishnan-Gehrke
Theory of Computation	Sipser; Ullman (Hopcroft-Motwani-Ullman)
Compiler Design	Aho-Lam-Sethi-Ullman (Dragon Book)
COA and Digital Logic	Hamacher; Patterson-Hennessy; Morris Mano
Discrete Maths	Kenneth Rosen; Liu
Probability and Statistics (DA)	Bertsekas and Tsitsiklis, Introduction to Probability
Linear Algebra (DA)	Gilbert Strang, Introduction to Linear Algebra
Machine Learning (DA)	Bishop; Hastie-Tibshirani-Friedman; Geron for practical intuition
Artificial Intelligence (DA)	Russell and Norvig, Artificial Intelligence: A Modern Approach
Python (DA)	Mark Lutz, Learning Python (if a refresher is needed)

- **PYQs:** solve topic-wise first, then full papers under timing. Aim to cover the last 10 to 15 years for CS and every available DA paper.
- **Test series:** use one good series for CS and one for DA; consistency beats variety.
- **Official material:** syllabus PDFs and any sample or mock paper from the GATE portal.

9. Exam-day strategy (3 hours, 65 questions)

Stage	Time	What to do
General Aptitude	About 20 min	Bank the 15 marks quickly and accurately
First pass, core	About 100 min	Answer every question you can do in under 3 minutes; mark long ones for review; attempt NAT and MSQ you know
Second pass	About 45 min	Return to marked questions; use elimination on MCQs; never leave NAT/MSQ blank
Final review	About 15 min	Check units, rounding on NAT, and MSQ option completeness

- For NAT, note the required rounding and answer format before typing.
- MSQ: one wrong option zeroes the question, but there is no negative marking - so pick only options you can defend.
- Do not chase one question; the 2-mark questions are worth the time only when you see the path.
- Because CS and DA are on different slots, treat them as two separate exams: sleep, travel and revise for each independently.

Common mistakes to avoid

- Skipping General Aptitude and Engineering Maths.
- Reading theory without solving PYQs and mocks.
- Treating DA as easy because it is shorter - depth in maths and ML is the difficulty.
- Starting mocks too late; begin full-length practice by late December at the latest.
- Neglecting the error log - repeated mistakes are the cheapest marks to recover.

10. After the exam

- Check the response sheet and answer key when released; note any challenge window.
- Results expected about 19 March 2027; the score is valid for 3 years, which keeps M.Tech/MS/PhD and PSU recruitment options open across several cycles.
- Admission to M.Tech programs and PSU shortlisting run on separate processes with their own portals and cut-offs; watch for them after results.

11. Sources and verification

Compiled from GeeksforGeeks GATE 2027 CS and DA syllabus pages, Made Easy, The ML Hub, Myndaq, PiyushAI and related coverage of the IIT Madras notification. Official syllabus PDFs:

gate2027.iitm.ac.in/static/doc/GATE2027_Syllabus/CS_GATE2027_Syllabus.pdf and [.../DA_GATE2027_Syllabus.pdf](https://gate2027.iitm.ac.in/static/doc/GATE2027_Syllabus/DA_GATE2027_Syllabus.pdf).

Weightage ranges and schedule suggestions are planning estimates, not official figures. Always confirm dates, fees, documents and rules on the official GATE 2027 portal.